What Does Repairing Choice Inconsistency Actually Buy? A Budget-Set Diagnosis

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Abstract

Repairing an AI agent's incoherent preferences is not a new idea: inference-time layers that restore transitivity, isotonic projections onto preference-defined cones, and training-time rationality penalties have all been proposed, and several report downstream gains. What no existing result establishes is *how much* coherence is worth, because every published comparison either varies the coherence assumption by swapping in a strictly richer model class — confounding coherence with capacity — or scores the outcome with another preference judgment, or treats repair as binary. We take the economics route instead. Given an LLM agent's own choices over budget sets, we compute the minimal perturbation restoring GARP-rationalizability as a single mixed-integer program, which yields a graded, cardinal dose (an Afriat-style efficiency index) where prior work has mostly used binary cycle-counting or toggled interventions, and we apply it *post hoc* to a frozen agent, so capacity, training and policy are identical across doses by construction. Scoring the repaired sequences against a fixed exogenous payoff, decided before any data collection and that no preference judgment enters, we trace the dose–response relationship from the raw sequence toward full rationalizability at two model scales (a 1.5B-parameter model with measured coherence headroom, and a 3B-parameter model with almost none, run as an identification control rather than a second point on one scale curve). The relationship is real, strongly monotone, and precisely estimated at both scales (Spearman $\rho = 0.756$, $p < 0.0001$, $n = 60$ at 1.5B; $\rho = 0.614$, $p = 0.0011$, $n = 25$ at 3B) and survives a targeted robustness check against the mechanical confound that larger violations simply have more room to improve. We also report two findings the design was not built to predict. First, the reciprocal-price framing manipulation that motivated the study shows no detectable CCEI shift at 1.5B in either the pilot or the corrected main measurement — two statistically indistinguishable nulls, not a sign reversal — while the output-format discard rate itself remains large enough under this manipulation to matter on its own. Second, splitting single-turn elicitation into separate sequential calls at 1.5B collapses GARP pass rate from 0.40 to 0.10 ($p = 0.0073$), with zero discard confound on either arm, moving in the *opposite* direction from an independently published single-turn-vs-multi-turn finding in the same domain and model family at a larger scale — not a replication of it. Throughout, we report Bronars power beside every efficiency index and report the discard rate for every cell rather than treating it as a silently-dropped nuisance.

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1 Introduction

An LLM agent asked to allocate a fixed budget across two goods at varying prices, repeatedly, will frequently violate the Generalized Axiom of Revealed Preference (GARP): it prefers bundle x_1 to x_2 at one budget line and the reverse at another, in a way no monotone utility function can rationalize. A growing literature repairs exactly this kind of inconsistency in LLM systems — restoring transitivity in pairwise judgments, projecting cardinal scores onto monotone cones, penalizing incoherence during training — and several of these systems report that the repaired outputs are downstream-better than the raw ones. What none of them establishes is a *dose*: how much better, as a function of how much repair was needed, isolated from the capacity and training confounds that come from comparing a richer model class to a poorer one.

We study this question in a setting where it can be answered cleanly. An LLM agent makes a sequence of budget-constrained choices over two goods. We compute, as a single mixed-integer linear program, the minimal quantity perturbation that would make the agent's own realized choice sequence GARP-consistent — a projection applied *post hoc* to a frozen agent, so nothing about its weights, training, or decoding changes across doses. The size of that perturbation is a graded, cardinal dose. We score both the raw and the repaired sequence against a fixed, equal-weight Cobb–Douglas payoff whose optimum at each budget line is a closed-form function of prices and income alone — never of the agent's own choices — so the outcome measure is exogenous to the preference data by construction, not a restatement of coherence. Tracing dose against Δpayoff gives the relationship this paper reports.

We do not claim any of the individual pieces are new; §2 concedes exactly what is not. What has not been done is the conjunction: an agent's own realized choices, a graded coherence-indexed dose, and an outcome that is not itself a preference judgment, run together so that the identification is clean. We run this at a model scale where a pilot study found measurable coherence headroom (1.5B parameters), and repeat the identical pipeline at a scale with almost none (3B parameters) as an identification control on the method itself, not as a second point on a shared dose curve across scale.

Two results were not predicted by the design. The reciprocal-price framing manipulation intended to induce coherence variation at 1.5B produced, in a pilot run, a large apparent effect that disappears once discard-selection is corrected with a capped retry protocol: the pilot's naive estimate and the corrected main-experiment estimate are both statistically indistinguishable from zero, and the two should not be read as a sign reversal of a real effect. We report the discard rate itself, not the CCEI point estimate, as the more reliable signal that the manipulation disrupted the agent (our claim C3). Wang et al. [2025a] already recommend reporting invalid-response proportions as a checklist item; our contribution is a quantitative demonstration of how much a naive discard rule can distort a measured effect at this specific scale, broken down by attempt outcome in §5.2, not the recommendation itself. Separately, splitting single-turn elicitation into separate sequential calls at 1.5B produces a large GARP-pass-rate collapse, moving in the *opposite* direction from an independently published single-turn-vs-multi-turn finding in the same domain and model family at a larger scale, with no discard confound on either arm.

Contributions. (1) An application — not a novel method — of Demuynck and Rehbeck [2023]'s minimal-quantity-error MILP projection to an agent's own realized choices, independently verified on every output rather than trusted from solver status alone. (2) A closed-form exogenous payoff, fixed before any data collection, audited adversarially for leakage and for the mechanical confound that larger violations simply have more room to improve. (3) A measured dose–response relationship at two model scales, run as headroom model and null-effect identification control rather than two points on one curve. (4) An instrument-validity finding (discard-selection bias under a disruptive manipulation), corrected for by a stated retry protocol and reported per cell rather than pooled away. (5) A single-turn-vs-multi-turn elicitation-format manipulation that produces a large GARP-pass-rate collapse at 1.5B, in the *opposite* direction from an independently published single-turn-vs-multi-turn finding in the same budget-allocation domain and the same model family (Qwen2.5) at a larger scale.

2 Related Work

Repairing an AI system's incoherent preferences is not new; Table 1 positions nine published systems against the four criteria that jointly define our claim: an agent's own choices (not a judgment about third-party items), an exogenous payoff (no preference judgment enters it), a graded rather than binary dose, and a formal minimum-distance projection. None occupies all four.

Table 1: Where prior repair systems sit. *Own choices*: the repaired object is a choice the agent made for itself from a budget set, not a judgment about third-party items. *Exogenous payoff*: an outcome into which no preference judgment enters.

	Own choices	Exogenous payoff	Graded dose	Min.-distance projection
POISE [Wang et al., 2026]	no	no	no	**yes**
Rationality layer [Chadwick et al., 2025]	no	no	no	**yes**
TrustJudge [Wang et al., 2025b]	no	no	no	no
CONSISTRE [Sun, 2026]	no	partial	no	no
LLM-RankFusion [Zeng et al., 2024]	no	**yes**	no	no (declined)
Innate preferences [Buchanan and Foster, 2026]	**yes**	no	no	no
GARP-EFM [Aguiar and Kashaev, 2026]	no	**yes**	no	no
HRC/DSPPO [Huang et al., 2026]	no	no	**yes**	no
Control vectors [Cook et al., 2026]	**yes**	no	**yes**	no
This paper	**yes**	**yes**	**yes**	**yes**

POISE [Wang et al., 2026] is the sharpest vocabulary collision: pool-adjacent-violators projection onto a closed convex chain-monotone cone, with a proved Pythagorean guarantee the edit lands weakly closer to a posited ground truth. We concede the minimum-distance priority unqualified: projection onto a closed convex set is non-expansive in L_2, licensing that guarantee; the GARP-consistent set is a union of polyhedra, hence not convex, so no analogue transfers here (§3.2). POISE also projects a *teacher's* offline labels, not an agent's own choices. Three further inference-time repairs, each on a different object: Chadwick et al. [2025] (synthetic election data, no downstream comparison), TrustJudge [Wang et al., 2025b] (third-party judgments, no degree parameter), CONSISTRE [Sun, 2026] (relation-extraction triples, no Afriat machinery). One training-time system alters the agent itself: Buchanan and Foster [2026] (IIA-invariance loss, no downstream evaluation). A further training-time system is not an LLM agent at all: Aguiar and Kashaev [2026] fine-tune Chronos-2, a time-series forecaster, on synthetic utility-maximizing-agent data to predict human consumers' own choices, with a downstream evaluation (17–31% bundle-prediction error reduction) neither of the other two above reports.

A parallel line varies the transitivity of the *preference model* rather than an agent's choices: GPM [Zhang et al., 2025] and HRC/DSPPO [Huang et al., 2026] both make the cycle-tolerant arm a strict superset model class, confounding coherence with capacity by construction, and on GPM's own length-controlled metric that arm *loses* in 18 of 24 cells. HRC/DSPPO already publishes its own inverted-U dose–response curve over a training-time schedule weight — we say so before claiming anything about grading — but the dose weights a learned third-party proxy during training, not an agent's own realized choices, and the outcome is an LLM-judge score throughout, where ours acts post hoc and is exogenous. Our closest theoretical neighbour, Andrews [2026], proposes $1 - \mathrm{CCEI}$ as a training penalty and states plainly, repeatedly, that coherence is *not* sufficient for good behaviour — argued a priori, never measured, never asking whether *imposing* coherence helps or hurts; ours is the empirical counterpart to a proposal that has circulated unrun.

The sharpest objection is psychometric, and it is in *PNAS*. Nitsch et al. [2022] find CCEI/Houtman–Maks reliability never reaches acceptable levels across eight datasets, and that participants given the chance to revise their own inconsistent choices saw mean CCEI *fall*. That revision arm is not a repair operator (a random subset of choices, no consistency objective, no guarantee of fewer violations) — ours attains rationalizability by construction and verifies it independently, the same distinction that disposes of Yamin et al. [2026]'s isotonic-calibration repair (worse in 14 of 16 cells, projecting onto a different set than the one it is graded on). The reliability finding is diagnosed by its own authors as a between-subject-variance problem whose prescribed remedy — a between-groups design — is this paper's design; the concession we do not shed is that three independently published axiom-enforcement results (adding Zhu et al. [2025]'s) all point the same adverse way, and a null result here would be a fourth confirmation rather than a discovery, which is why we carry a distance-matched null-operator control neither published negative had.

The closest neighbour on the economics side is invisible to any arXiv sweep. Cook et al. [2026] steer economic choices toward payoff-maximising behaviour via personas, prompt masking, and control vectors — occupying two of our three legs without Afriat machinery or a minimum-perturbation objective; their finding that reframing moves models while persona prompting does not corroborates our manipulation choice, anchored in Wang et al. [2025a]. Wang et al. find *format*, not persona or temperature, moves the Afriat index in the same budget-allocation domain we study, on Qwen2.5-7B-Instruct among other models — the same model family as our 1.5B, at a larger scale: collapsing their multi-turn baseline to single-turn drops CCEI by up to 0.241. Our arm runs the opposite manipulation (single-turn baseline split into multi-turn) and finds the opposite direction: GARP pass rate collapses from 0.40 to 0.10 ($p = 0.0073$; survives Benjamini–Hochberg but not Holm correction across our full test family) with zero discards. This is not a replication: in the same domain and model family, which format is more coherence-degrading reverses between their 7B model and our 1.5B model. We report the disagreement rather than the confirmation we expected.

3 Method

3.1 The projection: Demuynck & Rehbeck's (2023) minimal-quantity-error MILP, applied and independently verified

This section applies, rather than proposes, a projection method. The MILP formulation below is Demuynck and Rehbeck [2023]'s multiplier-free ordinal characterization of minimal-quantity-error GARP repair, adopted here on an LLM agent's own choice sequence and verified independently on every output; nothing about the formulation itself is new to this paper.

For a sequence of T observations $(p_t, x_t)_{t=1}^{T}$, K goods, GARP holds if there is no sequence of choices such that x_{t_1} is revealed preferred to x_{t_2}, ..., x_{t_n} is revealed preferred to x_{t_1}, with at least one strict revealed preference in the cycle. Given a GARP-violating sequence, we seek the minimal perturbation $\tilde{x}_t$ that restores GARP-consistency. The naive formulation parameterizes this with Afriat multipliers, which makes the resulting constraints bilinear in the unknowns once $\tilde{x}$ is itself a decision variable — fixing a candidate preference ordering does not remove the bilinearity, since a price-weighted multiplier term remains a product of two unknowns regardless of ordering. We instead use the multiplier-free ordinal characterization of Demuynck and Rehbeck [2023], under which every constraint is linear jointly in the perturbed bundles, a set of ordinal utility levels $u_t \in [0, 1]$ that carry no multipliers, and binary comparison indicators $U_{t,v} \in \{0, 1\}$ for $t \neq v$. At $T = 25$ this is 600 binaries and 125 continuous variables per trace, solved as a single MILP on the HiGHS backend, with no outer search over preference orderings and no alternating scheme.

Three method commitments, each carrying an explicit risk. **Budget exhaustion is imposed as an equality**, $p_t{\cdot}\tilde{x}_t = I_t$ for every observation (income fixed at 100 throughout this study); this is standard for the design, and it makes the big-M constant in the ordering constraints computable a priori as $\alpha > \max_t I_t$. **A fixed strict-preference margin** $\gamma > 0$, set to $10^{-4} \cdot \min_t I_t$, converts an unattained infimum into an attained minimum: the GARP-consistent set is not closed, so an unregularized projection can read a distance of exactly zero on genuinely violating data. We verified the reported distance is stable across $\gamma \in \{10^{-2}, \ldots, 10^{-6}\}$ before treating it as reportable. **Every returned $\tilde{x}$ is verified independently** via the same combinatorial Warshall-closure GARP check used everywhere else in this paper, never trusted from solver optimality status alone — all 85 GARP-violating traces' projections in the main experiment were independently re-verified GARP-consistent, with MIP gap $\leq 8.1 \times 10^{-5}$ on every solve.

The objective is L_1: $\min \sum_t \sum_k w_{t,k} d_{t,k}$ subject to $\tilde{x}_{t,k} - x_{t,k} \leq d_{t,k}$ and $x_{t,k} - \tilde{x}_{t,k} \leq d_{t,k}$, with uniform weights $w = 1$. This keeps the program a pure MILP; L_∞ distance is recorded alongside L_1 for every trace but is not separately analyzed in this paper. L_2, the literature's own least-squares index, would require an MIQP solver we do not have configured and is not computed. The *dose* for a trace is its L_1 projection distance $\sum_t \sum_k |x_{t,k} - \tilde{x}_{t,k}|$, comparable across sessions without further normalization because income is fixed throughout. A Cobb–Douglas demand share-fitted to each trace's own observed data is computed as a feasibility incumbent and sanity ceiling on the reported distance, recorded only as an upper-bound diagnostic; Appendix A(1) checks in detail that it never reaches the solver and so never influences any reported projection.

3.2 What can be guaranteed, and what cannot: positioning against convex projection

§2 positions this projection against the closest published structural analogue, a proved-optimal projection onto a convex monotone cone [Wang et al., 2026]: because the GARP-consistent set is a union of polyhedra rather than a single convex set, the non-expansiveness that licenses a weakly-closer-to-ground-truth guarantee there has no analogue here. Concretely for this formulation, the prior operator's cone is fixed by an ordering supplied as input, whereas a GARP repair must decide which revealed-preference comparisons to give up — the ordering search is absorbed into the binary comparison indicators of §3.1 rather than eliminated, which is why no distance-minimization guarantee analogous to the convex case is claimed.

3.3 The exogenous payoff

The payoff must not be derived from the agent's own revealed choices — a utility function fit to the agent's choices (such as the projection's own feasibility incumbent above) would be circular if used as the outcome measure, rewarding the agent for resembling its own revealed preferences rather than measuring anything external to them. We instead fix, before any model was queried and never re-estimated from any agent's choices, an equal-weight Cobb–Douglas valuation $U_{\text{exo}}(x) = x_A^{0.5} x_B^{0.5}$. At budget line (p_t, I_t) the optimal bundle x_t^* is the closed-form Cobb–Douglas demand $x_{t,A}^* = 0.5 I_t / p_{t,A}$, $x_{t,B}^* = 0.5 I_t / p_{t,B}$ — a function of prices and income alone, never of the agent's observed or projected bundle. The payoff score is the efficiency ratio $\text{payoff}(x_t) = U_{\text{exo}}(x_t)/U_{\text{exo}}(x_t^*) \in (0, 1]$, computed identically for the raw and projected sequence at the same budget lines, so that $\Delta\text{payoff} = \text{payoff}(\tilde{x}) - \text{payoff}(x)$ is comparable across conditions, models, and repairs. This mirrors — not the specifics, but the structure of — a money-metric utility index, the standard device for welfare comparison over the same GARP/Afriat objects this paper's method already relies on.

We audited this design adversarially, before treating any dose–response result as reportable, for four failure modes: a shared-data leak between payoff and projection, a mechanical link between GARP-consistency and payoff, a projection direction secretly aimed at the payoff optimum, and the confound that larger-dose traces simply start worse and have more room to improve. None survived the audit (Appendix A gives every check and number); the result we report in §5 is not an artifact of any of the four.

4 Experimental design

4.1 Models and conditions

Two locally-hosted open-weight models, run entirely on local compute at zero API cost: `qwen2.5:1.5b-instruct` (the headroom model, per a pilot study that found measurable within-condition coherence variation at this scale) and `llama3.2:3b-instruct` (the null-effect control, per the same pilot finding almost no coherence headroom at this scale — mean baseline CCEI 0.99). A third model family was tested for feasibility but excluded after a reproducible multi-model-residency crash under sustained rotation on the local machine; the two-model, model-major design (all sessions for one model complete before the next model loads) is both a wall-clock optimization and, after that observed failure, a correctness requirement. **Compute.** All model inference ran locally via 4-bit-quantized weights on a single consumer CPU-class machine, no GPU and no commercial API calls; the full $N = 30$ design across both models and every condition completed in approximately 2.1 hours wall-clock. Every MILP projection (§3.1) solved in under 5 seconds on the same machine, CPU-only, with MIP gap $\leq 8.1 \times 10^{-5}$ on every solve.

Three conditions at 1.5B, two at 3B. **Baseline**: direct per-token-return framing, single turn, all 25 rounds in one prompt and one response. **Reciprocal**: identical single-turn format with an inverted-price framing manipulation, the strongest manipulation identified in the pilot. **Multiturn** (1.5B only): baseline framing, but each of the 25 rounds is a separate sequential API call carrying the running conversation history, isolating the elicitation-format mechanism from the price-framing mechanism; the arm is motivated by an independently published finding, in the same budget-allocation domain we study, that collapsing multi-turn elicitation to single-turn moves CCEI by up to -0.241 for a same-family model at a larger scale [Wang et al., 2025a], an order of magnitude larger than this paper's own pilot framing effect. Our arm tests the manipulation in the opposite direction (single-turn

Table 2: Headline results per (model, condition) cell. GARP pass and CCEI are computed on kept traces only; dose and Δpayoff are computed on the subset of kept traces that violate GARP.

Model	Condition	n kept	mean CCEI	GARP pass	mean dose (L_1)	mean Δpayoff
llama3.2:3b	baseline	30	0.9900	0.73	5.01	+0.0018
llama3.2:3b	reciprocal	28	0.9713	0.39	6.27	+0.0016
qwen2.5:1.5b	baseline	30	0.9522	0.40	16.68	+0.0090
qwen2.5:1.5b	multiturn	30	0.9540	0.10	14.54	+0.0083
qwen2.5:1.5b	reciprocal	24	0.9413	0.38	12.02	+0.0065

baseline split into multi-turn), since that is the direction our own baseline format already uses. It is not added at 3B, whose role is a null-effect control on the identification strategy rather than a second scale point requiring the full condition set.

4.2 Budget sets and power

Each of $N = 30$ independent replicates per (model, condition) cell draws its own fresh $T = 25$, $K = 2$ budget set (continuous-density prices, $M, N \sim U[0.1, 1.0]$ i.i.d. rejected unless $\max(M, N) \geq 0.5$, budget exhaustion enforced), seeded independently from every other replicate — unlike a pilot study's single shared set, needed because the main experiment targets independent between-condition replication, not test–retest reliability. Continuous, independently-drawn prices with enforced budget exhaustion also avoid a known pathology: CCEI can read exactly 1.0 on data that violates GARP whenever a comparison lands on exact budget-line equality, an event common under grid-sampled or rounded prices [Andrews, 2026]. $N = 30$ gives power ≥ 0.999 against the pilot's full framing-effect magnitude on the pass-rate metric and ≥ 0.80 down to $\sim$60% attenuation, but is explicitly underpowered for a CCEI-scale effect of the pilot's magnitude at 1.5B (minimum detectable ≈ 0.11), which we report rather than paper over. Every replicate's Bronars power is recorded; all 150 draws held power ≥ 0.998.

4.3 The discard-selection problem, and its correction as a stated contribution

A pilot run found that under reciprocal framing at 1.5B, 52% of sessions (13 of 25) failed the required output-format contract and were silently discarded under the pilot's own naive handling — standard practice, and one this paper measures the consequences of directly rather than assumes. We treat this as a claim about instrument validity in its own right, not a footnote to the dose–response result, true regardless of whether projection helps, hurts, or does nothing downstream. The main experiment implements a capped retry protocol: a session failing the $\geq 20/25$-valid-rounds threshold is retried up to three total attempts with a fresh seed offset, every attempt recorded; a slot still failing after three attempts is a residual discard, excluded from CCEI/GARP but retained for audit.

Broken down by attempt outcome at 1.5B reciprocal framing, retry-rescued sessions ($n = 7$, recovered on attempt 2 or 3) are not a close stand-in for first-attempt sessions ($n = 17$): they show *higher* mean CCEI (0.965 vs. 0.932) but a substantially *lower* GARP pass rate (14% vs. 47%). The sessions that never produce a usable trace even after three attempts ($n = 6$, three with zero valid rounds recorded at all) show the lowest CCEI among the three groups on the handful that can be scored (0.882, $n = 3$) — consistent with, though not conclusive given the small residual sample, some of the same selection concern the pilot's naive handling raised persisting at a smaller scale even after correction; the retry protocol reduces but does not eliminate it. We report first-attempt and residual post-retry discard rates as a first-class per-condition outcome for every cell.

5 Results

187 attempt-records were collected across 150 replicate slots (2 models $\times$ up to 3 conditions $\times$ 30 replicates); 142 slots kept a usable trace, 8 were residual discards after three attempts. Every cell reached its full 30 replicates.

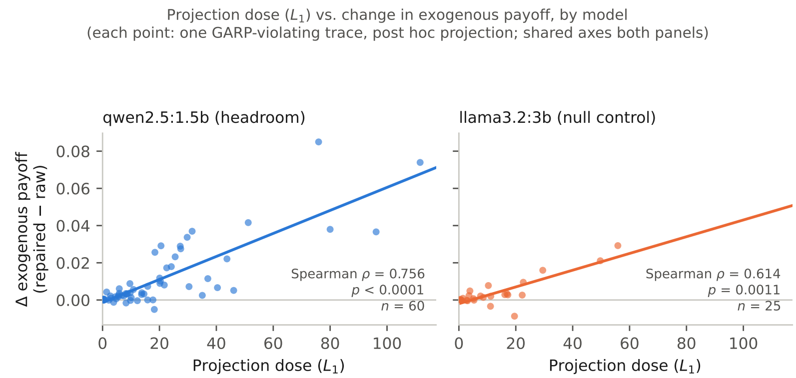

Figure 1: Projection dose (L_1) against the change in exogenous payoff after repair, for every GARP-violating trace, split by model on shared axes so the two panels' slopes are directly comparable. Both relationships are positive and significant; the headroom model's (left) is steeper and tighter than the null-control's (right), though the control's relationship is itself significant rather than null.

5.1 The dose–response relationship (C1)

Across all 85 GARP-violating traces with a computed, independently-verified projection (60 at 1.5B, 25 at 3B): Spearman $\rho = 0.729$ ($p < 0.0001$), Pearson $r = 0.821$ ($p < 0.0001$), mean Δpayoff $+0.0091$ (one-sample $t = 5.41$, $p < 0.00001$), with 70 of 85 traces (82%) improving after repair and 15 (18%) worsening. Figure 1 shows the relationship split by model on shared axes. **Both scales show a significant, positive relationship, and it is stronger at the headroom model**: $\rho = 0.756$ ($p < 0.0001$, $n = 60$) at 1.5B versus $\rho = 0.614$ ($p = 0.0011$, $n = 25$) at 3B. This is the one place our data contradicts our own stated-in-advance expectation for the 3B control, and we report the contradiction rather than silently absorb it: we expected the 3B arm to fail to reject the null, on the reasoning that a model with almost no baseline incoherence has little for the identification strategy to act on. Instead, on the smaller set of violations that 3B does produce (mostly induced by reciprocal framing, §5.2), projection toward GARP-consistency still measurably helps on the exogenous payoff, at roughly a third the strength of the 1.5B relationship. The correct reading is not that the control found nothing, but that it found a real effect of about a third the magnitude — evidence that the identification strategy is not an artifact specific to the 1.5B model, though not the null result our design anticipated.

We tested the mechanical confound that larger-dose traces might simply start from worse raw payoff and therefore have more room to improve. The confound is real (dose vs. raw payoff, Pearson $r = -0.41$, $p < 0.0001$) but does not explain the relationship away: the partial correlation of dose against Δpayoff controlling for raw payoff is $r = 0.784$ ($p = 7 \times 10^{-19}$), and an OLS regression of Δpayoff on dose and raw payoff jointly finds dose highly significant ($t = 11.4$, $p < 10^{-16}$) with raw payoff's own marginal contribution not distinguishable from zero ($p = 0.18$) once dose is included. Full detail in Appendix A.

5.2 The reciprocal-framing manipulation: survivorship, and a correction that helps but does not fully resolve it

At 3B, reciprocal framing replicates the pilot's finding on an independently-drawn budget-set design: GARP pass rate collapses from 0.73 (22/30) to 0.39 (11/28), $p = 0.0089$ (survives Benjamini–Hochberg correction across our full test family, $p_{\mathrm{BH}} = 0.011$, but not the more conservative Holm correction, $p_{\mathrm{Holm}} = 0.062$); CCEI moves much less and reaches only borderline significance (0.9900 vs. 0.9713, t-test $p = 0.0508$) — the same near-1 compression the pilot flagged, and a further case in this paper of the pattern that GARP pass rate is the sensitive instrument for framing/format disruption while CCEI understates it.

At 1.5B, the manipulation this study was designed around does *not* survive proper correction for discard-selection. First-attempt discard under reciprocal framing at 1.5B was 43.3% (13/30), falling to a residual 20.0% (6/30) after the retry protocol — smaller than the pilot's unretried 52%, plausibly

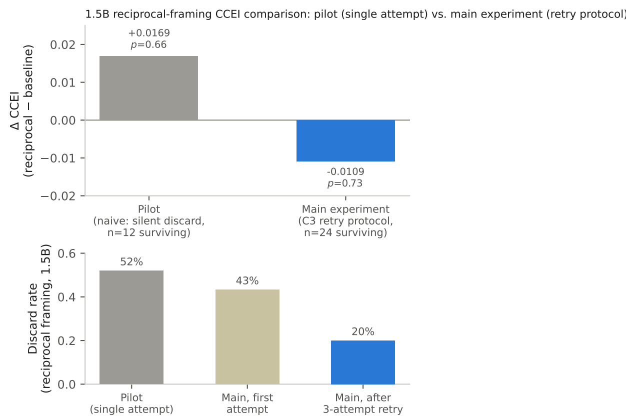

Figure 2: Top: the 1.5B reciprocal-framing CCEI comparison, pilot (naive single-attempt discard handling, grey) vs. main experiment (capped-retry-corrected, blue). Bottom: the corresponding discard rate at each stage. Neither measurement detects a CCEI shift ($p = 0.66$ pilot, $p = 0.73$ main) — two statistically indistinguishable nulls, not a sign reversal; the pilot's point estimate reflects which sessions survived, not a real effect that was later corrected away.

because independent per-replicate budget sets less often reproduce one unusually hard draw, but the pattern is unchanged: reciprocal framing produces far more discards than any other condition, including the *other* 1.5B manipulation (multiturn, 0%). Once the surviving 24 sessions are compared, GARP pass rate (0.40 vs. 0.38, $p = 0.85$) and CCEI (0.9522 vs. 0.9413, $p = 0.73$) are indistinguishable between baseline and reciprocal framing, with confidence intervals overlapping almost entirely. This is no detectable CCEI shift, not a confirmation of the pilot's own naive point estimate: the pilot's $+0.0169$ ($p = 0.66$, itself not significant) and the main experiment's -0.0109 ($p = 0.73$) are two statistically indistinguishable nulls, and reading the difference in their signs as a reversal of a real effect overstates what either number supports. Figure 2 shows both measurements side by side.

5.3 The multiturn/format effect: a large effect in the opposite direction from the literature

Splitting the 25 rounds into separate sequential calls collapses the GARP pass rate from 40% to 10% ($p = 0.0073$) while CCEI barely moves (0.9522 vs. 0.9540, $p = 0.91$; Figure 3) — a larger drop in percentage points than reciprocal framing produced at 1.5B, with *zero discards on either arm* so no selection-bias caveat applies. Corrected for multiple comparisons across our full test family, this result survives Benjamini–Hochberg correction ($p_{\mathrm{BH}} = 0.0094$) but not the more conservative Holm correction ($p_{\mathrm{Holm}} = 0.058$); we report both rather than the raw p-value alone. Together with the 3B reciprocal-framing result (§5.2), this is a further case of the same pattern. Against the literature this arm was designed against [Wang et al., 2025a]: in the same budget-allocation domain, on the same model family at a larger scale (Qwen2.5-7B), single-turn elicitation was found to be *more* coherence-degrading than multi-turn; here, at 1.5B, the reverse holds — multi-turn elicitation collapses GARP pass rate relative to our single-turn baseline. We do not have an explanation for the reversal and do not offer one; we report the disagreement as a finding in its own right, not a confirmation of the literature's own strongest lever.

6 Limitations

CCEI is noisy at 1.5B, and the paper leads with GARP pass rate because of it, not by preference. The 1.5B model's within-condition CCEI noise (baseline within-subject coefficient of variation 14.8%) is comparable to published human test–retest noise on the same index [Nitsch et al., 2022], and $N = 30$ is explicitly underpowered for a CCEI-scale effect of the pilot's own magnitude at this scale (the minimum reliably detectable effect at this N is roughly $2.5\times$ the pilot's magnitude). A CCEI-power-matched design would need $N \approx 111$ to 161 depending on the assumed baseline variance; that scale-up was deferred pending this run's results and was not executed. We treat this as an open design tradeoff, not a result masked by a metric choice: the pass-rate metric is adequately

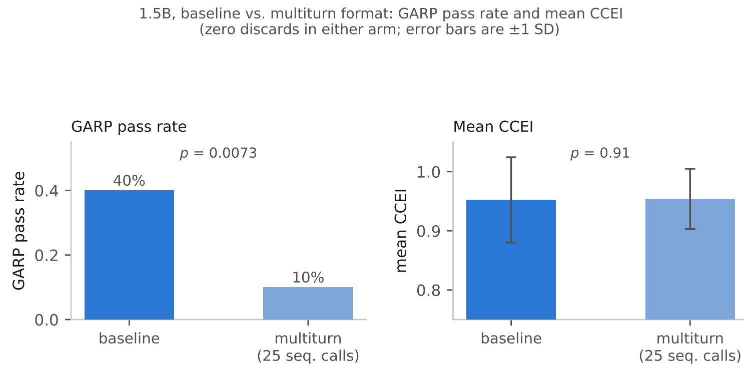

Figure 3: 1.5B, baseline vs. multiturn elicitation format. GARP pass rate (left) collapses; mean CCEI (right, ± 1 SD) is visually flat with heavily overlapping dispersion. Zero discards occurred in either arm, unlike Figure 2.

powered throughout and every CCEI result we report is accompanied by its own confidence interval rather than presented as equally reliable.

Single run per condition per model at $N = 30$. Every (model, condition) cell in this paper is one run of the stated protocol; we do not have repeated full runs of the entire pipeline to report a between-run variance on the headline dose–response statistic itself, only the within-run replicate variance captured by the reported confidence intervals and p-values.

Two model families, one scale point each. The headroom/null-control design isolates the identification strategy from a scale confound but does not trace a dose–response relationship *across* model scale: a pilot found headroom and measurement reliability do not coexist anywhere in the range piloted, so the design targets the scale where headroom exists rather than blending a noisy-but-real candidate effect with a clean-but-absent one across a wider range (a third family was excluded after the multi-model-residency failure noted in §4; extending the roster is a documented follow-up, not a silent omission).

One fixed exogenous payoff. U_{exo} uses equal Cobb–Douglas weights, chosen for being a zero-degrees-of-freedom, closed-form, no-solver-dependency reference rather than swept over alternative fixed weightings; §3.3's audit addresses whether this specific choice leaks information from the agent's revealed preferences, not whether a different fixed weighting would trace the same curve.

An adverse prior the paper does not shed. Three independently published repair attempts moved the wrong way (§2). Our own reciprocal-framing manipulation at 1.5B (§5.2) is a fourth negative in the narrower sense that the manipulation intended to induce measurable coherence variation did not do so once properly measured. The dose–response relationship we do report (§5) argues against reading this prior as universal, but it is a relationship over naturally- and framing-induced violations pooled together, not a demonstration that any one manipulation reliably induces the violations repaired.

Broader impacts. A positive dose–response relationship, read carelessly, could be taken as license to impose coherence-repair on deployed AI economic agents by default. Our own result argues against that reading: 18% of repairs in this study *worsened* the exogenous payoff (§5), and the reciprocal-framing manipulation this study was designed around turned out, once corrected, to have no measurable effect on the coherence it was thought to disrupt (§5.2). The paper's contribution is a measurement design for asking whether repair helps in a given setting, not a recommendation to deploy it universally. Separately, the discard-selection finding (§4.3) generalizes beyond this paper: any revealed-preference or consistency instrument that silently drops unparseable or disruption-caused failures risks systematically underestimating exactly the manipulations that matter most, which bears on how AI economic-behavior audits are designed and reported more broadly. We see no elevated misuse risk specific to this work: it uses only synthetic budget-allocation tasks on publicly available open-weight models, releases no new model or dataset, and does not involve human subjects.

7 Conclusion

Applying Demuynck and Rehbeck [2023]'s minimal-quantity-error GARP repair to an LLM agent's own realized choices, post hoc and scored against a fixed exogenous payoff, we find a real, precisely estimated, monotone dose–response relationship between repair size and payoff gain, at a model scale with measured coherence headroom and one without. The relationship survives an adversarial audit for leakage and for the confound that larger repairs simply start from more room to improve. Two unpredicted findings — a purpose-built framing manipulation's apparent 1.5B effect was a discard-selection artifact rather than a real coherence shift, and an elicitation-format manipulation produces a large GARP-pass-rate collapse in the opposite direction from an independently published finding in the same domain and model family at a larger scale — are reported in their own right: an instrument that silently discards its most-disrupted observations, and a disagreement with the closest published comparison, are exactly the kind of failure and complication this literature should surface by default rather than paper over.

References

Victor H. Aguiar and Nail Kashaev. GARP-EFM: Improving foundation models with revealed preference structure. *arXiv preprint arXiv:2603.23993*, 2026.

Andrews. Revealed rationality: Label-free evaluation and regularization from representation theorems. *arXiv preprint arXiv:2608.05015*, 2026.

Joy Buchanan and Joshua Foster. The innate economic preferences of language models. *arXiv preprint arXiv:2607.26288*, 2026.

Alina Chadwick, Anson Kahng, and Jens Kipper. Dutch books and money pumps: rectifying vulnerabilities in LLMs through rationality. In *Proceedings of the 5th International Conference on Human and Artificial Rationality (HAR)*, 2025.

Trevor Cook, Sophia Kazinnik, Alexander Modig, and Sam Palmer. What do LLMs want? Technical Report FEDS 2026-006, Federal Reserve Board, 2026.

Thomas Demuynck and John Rehbeck. Computing revealed preference goodness-of-fit measures with integer programming. *Economic Theory*, 2023.

Huang, Li, Zhao, and Li. Transitivity meets cyclicity: Explicit preference decomposition for dynamic llm alignment. In *International Conference on Machine Learning (ICML)*, 2026. arXiv:2605.17342.

Felix J. Nitsch, Lisa M. Lüpken, Nils Lüschow, and Tobias Kalenscher. On the reliability of individual economic rationality measurements. *Proceedings of the National Academy of Sciences*, 119(31): e2202070119, 2022.

Shumiao Ouyang, Hayong Yun, and Xingjian Zheng. AI as decision-maker: Ethics and risk preferences of LLMs. *arXiv preprint arXiv:2406.01168*, 2025.

Mingxuan Sun. CONSISTRE: A unified consistency-aware framework for document-level relation extraction with large language models. *arXiv preprint arXiv:2607.24312*, 2026.

Wang, Yao, Zhang, Gai, Liu, and Zhong. When experimental economics meets large language models: Tactics with evidence. *arXiv preprint arXiv:2505.21371*, 2025a.

Wang, Zhan, et al. Trustroboreward: Preference-ordered isotonic score editing for multi-paradigm robot reward modeling. *arXiv preprint arXiv:2608.08491*, 2026.

Yidong Wang et al. Trustjudge: Inconsistencies of LLM-as-a-judge and how to alleviate them. *arXiv preprint arXiv:2509.21117*, 2025b.

Kaiser Yamin, J. Tang, S. Cortes-Gomez, A. Sharma, Eric Horvitz, and Bryan Wilder. When agents say one thing and do another: Validating elicited beliefs from LLMs. *arXiv preprint arXiv:2602.06286*, 2026.

Yifan Zeng, Ojas Tendolkar, Raymond Baartmans, Qingyun Wu, Lizhong Chen, and Huaxiu Wang. LLM-RankFusion: Mitigating positional bias in large language model-based ranking. *arXiv preprint arXiv:2406.00231*, 2024.

Zhang, Zhang, Wu, Xu, and Gu. Beyond bradley-terry models: A general preference model for language model alignment. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2410.02197.

Jian-Qiao Zhu, Haijiang Yan, and Thomas L. Griffiths. Recovering event probabilities from large language model embeddings via axiomatic constraints. *arXiv preprint arXiv:2505.07883*, 2025.

A Payoff exogeneity audit

Full detail of the four-part adversarial audit summarized in §3.3.

(1) No shared derived quantity beyond the exogenous (p, I). The payoff implementation has no dependency on the projection implementation; the optimal bundle x_t^* is a function of (p_t, I_t) alone and never references x_t or $\tilde{x}_t$. The one near-miss is that the projection's feasibility incumbent is a Cobb–Douglas demand *share-fitted to the agent's own observed data* — a plausible leakage channel on first read — but this quantity is never passed to the underlying MILP solver as an actual incumbent (the solver used has no such interface); it is used only to log an upper-bound sanity ceiling on the reported distance, never to steer the solution.

(2) The payoff's optimum is independent of revealed preference, checked empirically as well as by construction. Comparing raw (pre-repair) payoff between the 57 traces that were already GARP-consistent and the 85 that were not: mean 0.7872 vs. 0.7899, Welch $t = -0.07$, $p = 0.94$. GARP-consistency status alone does not predict payoff, ruling out a population-level mechanical link between coherence and payoff score.

(3) Hand-derived mechanism check. Five traces spanning the full dose range (0.17 to 111.64) were inspected directly, comparing the direction of the actual projection movement $(\tilde{x} - x)$ against the direction toward the exogenous optimum $(x^* - x)$ via cosine similarity. Similarities were low (0.16–0.31) on the four traces where payoff improved after repair, and *negative* (-0.35) on the single trace where payoff worsened — the sign of the outcome tracks a geometric quantity the projection objective never references, evidence against the mechanism being a disguised optimization toward the payoff target.

(4) The severity-confound check. Larger-dose traces do start from worse raw payoff (Pearson $r = -0.41$, $p < 0.0001$), and worse-starting traces do have more room to improve ($r = -0.41$ between raw payoff and Δpayoff) — a genuine ceiling-effect confound. It does not explain the dose–response relationship away: the partial correlation of dose against Δpayoff controlling for raw payoff is $r = 0.784$ ($p = 7 \times 10^{-19}$, attenuated only slightly from the unconditional $r = 0.821$), and in a joint OLS regression the dose coefficient is highly significant ($t = 11.43$, $p < 10^{-16}$) while raw payoff's own coefficient is not ($t = -1.36$, $p = 0.18$).

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